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ORIGINAL ARTICLE

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Genetic variation and the relationship between stream and lake ecotypes of a threatened desert Catostomid, the Warner sucker (*Catostomus warnerensis*)

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Abstract

Many fish species exhibit diverse life history strategies that help maintain population viability. An understanding of the relationships among these strategies is crucial for prioritising conservation actions. The Warner sucker, endemic to the Warner Lakes Basin in southern Oregon, USA, is one example of a taxon where a lack of information regarding relationships among life history strategies has hampered conservation efforts. Warner suckers have two distinct life history types: stream-type fish that have a fluvial life history and lake-type fish that have an adfluvial life history. There are advantages and disadvantages associated with each life history, and presently the relationship between life history types is not well understood. Our objectives were to determine the amount of genetic variation within and among tributary populations of Warner suckers and to determine the origins of suckers collected in the Warner Lakes. We collected individuals from four tributary populations, a refugial population, and Hart and Crump lakes and genotyped them at 16 microsatellite loci. Estimates of genetic variation among populations suggested low levels of gene flow (F_{ST} = 0.153) and genetic variation among populations seemed to be influenced by population and habitat characteristics. Nearly all of the individuals collected in Hart and Crump lakes originated in a single tributary, Deep Creek, which likely reflects reduced habitat connectivity between most other tributaries and the Warner Lakes. Data presented in this study are useful for evaluating the status of Warner sucker populations and for prioritising conservation actions such as the removal and modification of barriers.

KEYWORDS

Warner sucker, genetics, microsatellites, life history variation, genetic assignments, population introductions

1 | INTRODUCTION

Many fish species have evolved diverse life history strategies in order to maintain population viability. Examples of life history diversity include diversity in migratory patterns (Balazik, Garman, Van Eenennaam, Mohler, & Woods, 2012; Brenkman, Corbett, & Volk, 2007; Moore, Yeakel, Peard, Lough, & Beere, 2014; Northcote, 1997), diversity in age at maturity (Moore et al., 2014; Schindler et al., 2010), variation in spawning frequency (Auer, 1999; Brenkman et al., 2007) and diversity in age structure of populations (Hellmair & Kinziger, 2014; Schindler et al., 2010). This so-called portfolio effect (Figge, 2004), where risks are spread among multiple life history types and cohorts, helps to buffer populations against changing environmental conditions, stochastic events, predation and exploitation (Greene, Hall, Guilbault, & Quinn, 2010; Hilborn, Quinn, Schindler, & Rogers, 2003; Schindler et al., 2010). In order to conserve species

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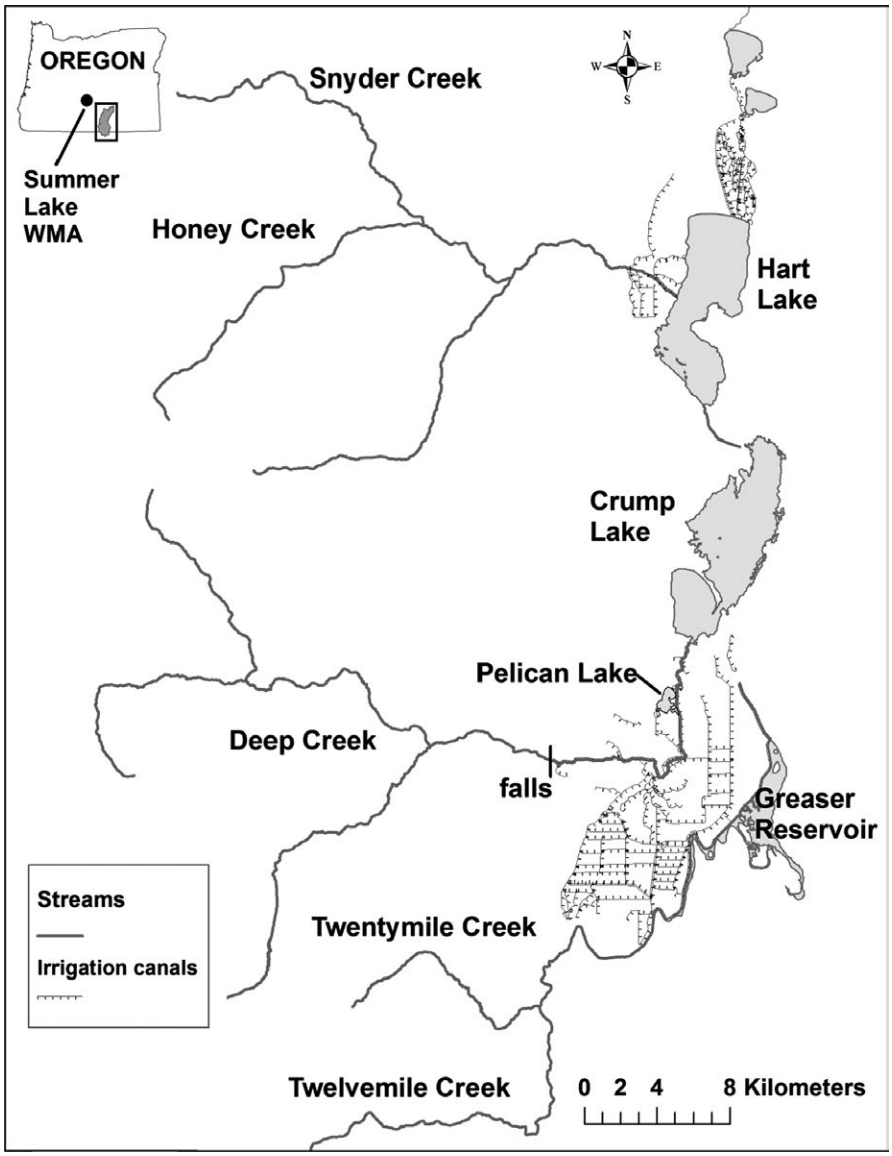

FIGURE 1 Study location, the Warner Basin, Oregon, USA. Warner suckers were collected from Twentymile, Deep, Honey and Snyder creeks as well as Hart and Crump lakes. Individuals were also collected from a refugial population in the Summer Lake Wildlife Management Area (WMA). The box on the inset map shows the location of the Warner Basin within the state of Oregon

with variable life histories, it is important that biologists and fisheries managers have an understanding of the unique requirements of the different life history types and how they interact to maintain population viability.

Effective conservation plans for these species typically require a variety of management actions spread across different habitat types and over different time periods. For example, many fish species have both stream resident and migratory life history types (Northcote, 1997). Conservation for stream resident fish may depend on maintaining favourable stream flows and temperatures (Muhlfeld et al., 2012; Petty, Hansbarger, Huntsman, & Mazik, 2012; Pierce, Podner, & Carim, 2013), whereas conservation of migratory fishes may be more dependent on maintaining migratory corridors and removing barriers (Duda, Freilich, & Schreiner, 2008; Hitt, Eyler, & Wofford, 2012; Pess, McHenry, Beechie, & Davies, 2008; Shepard, May, & Urie, 2005). Much of our understanding of life history diversity in fishes is based on studies of commercially and recreationally important salmonid species. However, life history diversity can be found in many fish species spanning a broad range of families, and many of these species have very different life histories and are found in very different environments than salmonids. Conservation plans will require species-specific data to adequately identify and address the requirements for maintaining existing life history diversity and to ensure long term population viability.

The Warner sucker (*Catostomus warnerensis*) is an endemic Catostomid found in the Warner Basin, an arid sub-basin of the Great Basin in south-eastern Oregon and north-western Nevada, USA. Warner suckers inhabit the lakes and low-gradient stream reaches of the Warner Basin and the species has two distinct life history types: lake and stream morphs. The lake suckers exhibit a lacustrine–adfluvial pattern in which they spend most of the year in large lakes and presumably return to spawn in tributary streams. However, when upstream migration is hindered by low stream flows during drought years or by irrigation diversion dams, lake suckers may spawn in nearshore areas of the lakes (White, Ramsey, Stern, & Munhall, 1990). The stream suckers exhibit a fluvial life history

pattern and spawn in major tributary drainages to the Warner Lakes. There are advantages and disadvantages associated with each life history type. Warner suckers in the lakes have access to a greater forage base and can attain larger sizes, which presumably results in greater fecundity. However, the Warner Lakes have large populations of introduced fishes that reduce recruitment of Warner suckers by preying upon their young (U.S. Fish and Wildlife Service [USFWS] 1998). Periodic lake desiccation also threatens the lake suckers. Stream-type suckers inhabit a more stable environment with reduced risk of predation. However, resources are more limited in the streams, and threats specific to the stream type include water withdrawals for irrigation and habitat degradation associated with grazing (USFWS 1998). Presently, the relationship between the two life history types is not well understood. It is presumed that suckers in the Warner Lakes originate mostly in tributaries, but the tributary of origin for these individuals is unknown. Only limited movements of suckers between the lakes and the tributaries have been observed, and these are limited to observations of suckers which had been marked with passive integrated transponder (PIT) tags in the lakes being later detected in the lower reaches of the tributaries (Scheerer, Clements, Jacobs, & Peterson, in press).

Although historically abundant, Warner suckers have declined substantially due to the impacts of predation by non-native fishes and the construction of irrigation structures in the tributaries (USFWS 1985). As a result of these declines and the persistence of threats to the species, the Warner sucker was listed as "threatened" under the U.S. Endangered Species Act in 1985 (USFWS 1985). A greater understanding of population viability and the relationship between the two life history types is important for developing effective conservation and management plans to help recover the species. The Recovery Plan for the Threatened and Rare Native Fishes of the Warner Basin and Alkali Subbasin (USFWS 1998) outlines three criteria that must be met before delisting the species. These criteria require that (i) a self-sustaining metapopulation is distributed throughout the drainages of Twentymile Creek, Honey Creek and below the falls on Deep Creek, and in Pelican, Crump and Hart lakes; (ii) passage is restored within and among these drainages so that individual populations of Warner suckers can function as a metapopulation; and (iii) no threats exist that would likely threaten the survival of the species over a significant portion of its range.

Biologists recognise the importance of restoring and maintaining connectivity among lake and stream habitats so that Warner suckers can fully express the diversity of life history types necessary for maintaining a functioning metapopulation. In this study, we used genetic data to assess the status of Warner sucker populations and the relationship between the two life history types. The specific objectives of this study were twofold. First, we wished to determine the amount of genetic variation that exists within and among tributary populations of Warner suckers. This information will be important for inferring the viability and conservation status of these threatened populations and for determining the amount of genetic exchange among populations. Our second objective was to examine the relationship between lake and stream life history type Warner suckers. A clear understanding of this relationship is crucial for prioritising conservation actions in the Warner Basin such as the removal or modification of barriers which limit movements between stream and lake habitats.

1.1 | Study area

The Warner Valley is a north-east to south-west trending endorheic basin in Oregon, USA, that extends approximately 90 km (Fig. 1). The elevation of the valley floor is approximately 1,370 m and the basin is bound by fault block escarpments, the Warner Rim on the west and Hart Mountain and Poker Jim Ridge on the east. The Warner Basin was formed during the middle Tertiary and late Quaternary geologic periods as a result of volcanic and tectonic activity (Baldwin, 1974). Abundant precipitation during the Pleistocene Epoch resulted in the formation of Pluvial Lake Warner (Hubbs & Miller, 1948). At its maximum extent approximately 11,000 years ago, the lake reached approximately 100 m in depth and 1,300 km^2 in area (Snyder, Hardman, & Zdenek, 1964; Weide, 1975). Presently, the basin consists of three permanent lakes, Hart, Crump and Pelican, as well as several ephemeral lakes, and several tributaries. The basin is characterised by relatively low annual precipitation, and as a result, water levels in the permanent lakes vary widely among years and periodically the lakes completely desiccate. Streams in the Warner Basin are fragmented by an extensive network of irrigation diversions and canals that divert water for agricultural purposes (Fig. 1).

There are five fish species native to the Warner basin: Warner suckers, tui chub (*Siphateles bicolor*), redband trout (*Oncorhynchus mykiss*), Northern (Pit) Roach (*Lavinia mitrulus*) and speckled dace (*Rhinichthys osculus*). The lakes also contain an abundance of introduced, non-native fishes including white crappie (*Pomoxis annularis*), black crappie (*Pomoxis nigromaculatus*) and brown bullhead (*Ameiurus nebulosus*) that prey extensively on native species. Within the basin, Warner suckers are presently found in Hart and Crump lakes, and in the Honey, Deep and Twentymile creek drainages. Recent surveys in the Warner Basin tributaries documented suckers to be abundant and widely distributed (Scheerer et al., in press). Surveys in the Warner Lakes documented mostly older suckers and recent recruitment appears to be limited to a single year class (Scheerer et al., in press). A refugial population of Warner suckers also exists in an irrigation canal in the adjacent Summer Lake Basin (Fig. 1). This population was established at the Summer Lake Wildlife Management Area (WMA) in 1991 during a drought that desiccated the Warner Lakes (White, Ramsey, Stern, & Munhall, 1991). Seventy-five adult Warner suckers were temporarily translocated from Hart Lake to the irrigation canal and then subsequently moved either to Dexter National Fish Hatchery in New Mexico or back into the Warner Basin following the drought. Prior to being moved out of the irrigation canal, adult suckers spawned and the offspring were the first generation of the current Summer Lake WMA population. Although the number of adults that initially spawned in the Summer Lake WMA is unknown, the translocated adults were believed to be roughly 70% females and 30% males (White et al., 1991).

2 | METHODS

2.1 | Sample collection

We used backpack electrofishers to collect suckers from Twentymile, Deep, Honey and Snyder creeks and from the Summer Lake WMA irrigation canal. We collected suckers from 2006 to 2012 during studies designed to assess the current distribution, abundance and movements in the basin (Richardson et al., 2009; Scheerer, Heck, Gunckel, & Jacobs, 2007; Scheerer, Jacobs, Bratcher, Swearingen, & Kramer, 2008; Scheerer, Jacobs, & Munhall, 2006; Scheerer, Jacobsen, et al., 2011; Scheerer, Jacobs, et al., in press; Scheerer et al., in press). In Honey and Deep creeks, suckers were collected during a single year (2007) and collections were made over multiple years in Snyder Creek, Twentymile Creek and the Summer Lake WMA canal (Table 1). Suckers collected from tributaries ranged in size from approximately 50–335 mm. We used trap nets to capture Warner suckers in Hart and Crump lakes. We sampled the lakes during April, May and June of 2006, 2008 and 2012. In 2010, we only sampled Crump Lake due to low water depth in Hart Lake (<0.5 m). Suckers captured in the lake ranged in size from 50 to 450 mm.

2.2 | Laboratory methods

We extracted DNA from all individuals using QIAGEN DNeasy 96 Blood & Tissue Kits (Qiagen Inc., Valencia, CA, USA). We then genotyped all individuals at 16 microsatellite loci previously developed for other Catostomid species (Table 2). We conducted polymerase chain reactions (PCRs) in 10 μl volumes containing 2 μl of template DNA, 5 μl of 2x Qiagen multiplex PCR master mix (final concentration of 3 mM $MgCl_2$) and 0.2 μl of oligonucleotide PCR primer mix. Final

TABLE 1 Sample sizes and collection years for Warner suckers analysed for this study.

	Collection year						
Collection location	2006	2007	2008	2009	2010	2012	Total
Honey Creek		44					44
Snyder Creek		16				3	19
Deep Creek		26					26
Twentymile Creek		24	11	27	13		75
Summer Lake WMA		33		35			68
Crump Lake	2		26		30	34	92
Hart Lake	13		74			145	232

TABLE 2 List of microsatellite loci and PCR conditions used in this study. The locus *US4* was not part of a multiplex reaction.

Locus name	Multiplex name	Annealing temperature	Final primer concentration (μM)	Allele size range	Source
RBS04	Cwa4a	54	0.10	275–425	W. Wilson, USFWS, personal communication 16 March 2015
RBS20	Cwa4a	54	0.10	225–350	W. Wilson, USFWS, personal communication 16 March 2015
Xte1	Cwa4a	54	0.05	125–200	Turner et al. (2009)
Xte10	Cwa4a	54	0.10	275–325	Turner et al. (2009)
Dlu4105	Cwa5	57	0.10	125–300	Tranah, Agresti, & May (2001)
Dlu416	Cwa5	57	0.10	125–400	Tranah et al. (2001)
Dlu4243	Cwa5	57	0.10	80–175	Tranah et al. (2001)
Xte4	Cwa5	57	0.05	275–300	Turner et al. (2009)
Dlu26	Cwa6	52	0.10	150–375	Tranah et al. (2001)
Dlu4184	Cwa6	52	0.05	150–300	Tranah et al. (2001)
Xte12	Cwa6	52	0.05	150–225	Turner et al. (2009)
Xte6	Cwa6	52	0.05	325–400	Turner et al. (2009)
Dlu4183	Cwa7	52	0.05	125–275	Tranah et al. (2001)
Dlu45	Cwa7	52	0.10	275–425	Tranah et al. (2001)
RBS23	Cwa7	52	0.10	225–425	W. Wilson, USFWS, personal communication
US4		52	0.05	150–325	Cardall, Bjerregaard, & Mock (2007)

primer concentrations and annealing temperatures for each multiplex are listed in Table 2. Thermal profiles consisted of initial denaturation at 95°C for 15 min, followed by 30 cycles of denaturing at 95°C for 30 s, 1.5 min at multiplex-specific annealing temperatures (Table 2), extension at 72°C for 1 min and then a final extension at 60°C for 20 min. Following PCR, we conducted capillary electrophoresis on an ABI 3130xl Genetic Analyzer (Life Technologies, Foster City, CA, USA) following the manufacturer's protocols. We used the G5 filter set to produce electropherograms, and we analysed electrophoresis data using the program GeneMapper v4.0 (Applied Biosystems).

2.3 | Statistical analysis

We grouped Warner suckers collected from tributaries according to their collection location (Deep Creek, Honey Creek, Twentymile Creek, Summer Lake WMA) for statistical analysis. Collection efforts in the Honey Creek drainage included individuals from both Honey Creek and Snyder Creek (a tributary to Honey Creek). Preliminary analysis revealed that Honey and Snyder creeks represented separate populations and therefore we split individuals from these two tributaries into separate populations for analysis (Appendix S1). Based on tests for genetic variation among temporal replicate samples from Twentymile Creek and the Summer Lake WMA canal, we chose to combine temporal replicate samples from these tributaries for analysis (Appendix S2). We assumed that Warner suckers collected in Hart and Crump lakes were a mixture of individuals from the various tributary populations and we used these individuals for genetic population assignment analyses. We tested tributary populations for conformance to Hardy–Weinberg equilibrium (HWE) and for evidence of linkage disequilibrium (LD) using the program GENEPOP v4.1 (Raymond & Rousset, 1995). We adjusted significance values for HWE and LD tests using a sequential Bonferroni adjustment (Rice, 1989).

We estimated measures of genetic variation within each population including the mean number of alleles per locus (A_M), expected heterozygosity (H_{exp}) and observed heterozygosity (H_{obs}) for each of the five tributary collections using the program GDA (Lewis & Zaykin, 2001). We also calculated estimates of allelic richness (A_R) for each population using the program HP-Rare (Kalinowski, 2005). This program provides estimates of allelic richness corrected for differences in sample size among the collections. We calculated allelic richness based on a minimum sample size of 38 genes. We tested tributary collections for evidence of a recent genetic bottleneck using the program BOTTLENECK (Piry, Luikart, & Cornuet, 1999). This method tests for an excess of heterozygotes relative to the frequency of alleles in the population (Cornuet & Luikart, 1996). We assumed a two-phased model of mutation with 95% stepwise mutations and 12% variance as suggested by Piry et al. (1999). We used a one-tailed Wilcoxon test to determine whether collections showed evidence of a significant genetic bottleneck.

Oftentimes introduced populations show reduced levels of genetic variation and/or increased relatedness when compared to wild source populations (Mock, Latch, & Rhodes, 2004; Puckett et al., 2014; Stephen, Whittaker, Gillis, Cox, & Rhodes, 2005). We compared levels of genetic variation observed in the introduced Summer Lake WMA population to levels of variation in natural populations of Warner suckers to determine whether the introduced population had levels of variation comparable to natural populations. We used Wilcoxon ranked sum tests to determine whether there were significant differences in estimates of A_R and H_{exp} between Summer Lake and the four natural origin tributary populations. We also calculated two separate estimates of pairwise relatedness within each population to determine whether levels of relatedness in Summer Lake were equivalent to those in the natural populations. We first estimated Queller and Goodnight's (1989) coefficient of relatedness (r_{xy}) for all pairs of individuals in each population using the program SPAGeDi (Hardy & Vekemans, 2002). This coefficient ranges from −1.0 to 1.0 with values greater than zero indicative of two individuals with above average relatedness (i.e. full and half siblings) and values less than zero indicative of two individuals with below average relatedness. Next, we estimated pairwise *r* values using the maximum-likelihood approach implemented in the program MLRelate (Kalinowski, Wagner, & Taper, 2006). Values of *r* range from 0.0 to 1.0 with pairwise *r* values of approximately 0.5 representing two individuals that are full siblings or parent–offspring pairs, pairwise *r* values of approximately 0.25 representing two half siblings and pairwise *r* values of 0.0 representing two unrelated individuals. We compared the distribution of both relatedness estimators among the five populations.

We estimated the overall level of genetic variation among tributary collections (F_{ST}) and the associated 95% confidence interval (based on 1,000 bootstrap replicates) using the program FSTAT v2.9.3.2 (Goudet, 2001). We also used FSTAT to estimate the level of genetic variation among all pairs of tributary collections (pairwise F_{ST}). We used GENEPOP to conduct contingency tests of allele frequency heterogeneity to determine whether there were significant differences in allele frequencies among the different tributary collections. Significance values for allele frequency heterogeneity tests were adjusted for multiple comparisons using a sequential Bonferroni adjustment. To examine the relationship among populations, we conducted a multivariate discriminant analysis of principal components (DAPC; Jombart, Devillard, & Balloux, 2010) of our allele frequency data using the *adegenet* package (Jombart, 2008) for the R statistical environment (R Development Core Team 2015). DAPC is similar to principle component analysis (PCA) but unlike PCA, which maximises the total variation in the data set, DAPC maximises the variation among different groups and minimises variation within groups (Jombart et al., 2010).

We assumed that Warner suckers collected in the lakes originated in the tributaries and we used genetic population assignments to determine the origins of these fish. In order to determine our ability to assign unknown origin individuals from the lakes to their tributary of origin, we first performed leave-one-out assignment tests of our baseline data set using the program GeneClass2 (Piry et al., 2004). For this analysis, individuals are removed from the baseline data set one at a time, the baseline allele frequencies are then recalculated without that individual and the unknown origin individual is assigned to its most likely population of origin. The number of individuals assigned to the tributary they were collected from provides an assessment of the accuracy of the baseline data set for assigning unknown origin fish to

their most likely population. This analysis relies on the assumption that the baseline fish were collected in their tributary of origin and were not migrants from another tributary. We conducted leave-one-out assignment tests using a baseline data set that consisted of individuals from Deep, Honey, Snyder and Twentymile creeks. The Summer Lake WMA collection was excluded from the baseline since Summer Lake is a separate drainage and is not connected to the Warner Lakes Basin. Genetic assignments and the associated likelihood scores (ranging from 0.0 to 100.0) for individual assignments in the leave-one-out tests were determined using the methods of Rannala and Mountain (1997). Following leave-one-out tests, we assigned unknown origin Warner suckers collected in Hart and Crump lakes from 2006 to 2012 to their most likely population of origin using GeneClass2.

3 | RESULTS

3.1 | Genetic variation within and among tributary populations

Prior to Bonferroni correction, we observed three loci that deviated from HWE expectations in Honey Creek and two loci that deviated from HWE expectations in the introduced Summer Lake population. Following Bonferroni correction, we did not observe any deviations from HWE expectations. We observed the following number of locus pairs (out of 600 total) that showed evidence of linkage: Deep Creek—three locus pairs; Honey Creek—three locus pairs; Snyder Creek—zero locus pairs; Twentymile Creek—six locus pairs; Summer Lake WMA—14 locus pairs. We observed no clear patterns of linkage among loci; therefore, we retained all loci for subsequent analysis.

Estimates of genetic diversity did not vary substantially among the five locations (Table 3). All estimates of diversity were lowest in Honey Creek (A = 6.56, A_R = 5.78, H_{exp} = 0.669 and H_{obs} = 0.653) and all estimates of genetic diversity were greatest in Deep Creek (A = 8.25, A_R = 7.70, H_{exp} = 0.775 and H_{obs} = 0.811). Only Twentymile Creek showed evidence of a recent genetic bottleneck (Wilcoxon test $p < 0.001$). When we compared estimates of allelic richness and expected heterozygosity in the introduced Summer Lake population to the four natural populations, both estimates were significantly greater in Deep Creek (Wilcoxon $p = 0.003$ and 0.021, respectively) and we observed no significant differences for any of the other comparisons ($p > 0.05$). The distributions of both pairwise relatedness estimators were similar between Summer Lake and all natural populations (Fig. 2).

The overall level of variation among all sampling locations (global F_{ST}) was 0.153 and was significantly different from 0.0 (95% C.I. = 0.124–0.186). Pairwise estimates of F_{ST} ranged from 0.082 for the comparison between Deep and Honey Creek to 0.212 between Honey Creek and Summer Lake (Table 4). The Summer Lake population was most genetically similar to Deep Creek based on pairwise F_{ST} estimates (Table 4). Tests of allele frequency heterogeneity indicated that there were significant differences in allele frequencies among nearly all loci for each population pair (Table 4). Principle components 1 and 2 in the DAPC separated individuals into four clusters corresponding to Twentymile Creek, Summer Lake, Deep Creek, and one cluster consisting of individuals from Honey Creek and Snyder Creek (Fig. 3). The third principle component in the DAPC separated individuals from Honey and Snyder creeks into two distinct clusters (Fig. 3).

TABLE 3 Estimates of genetic diversity based on 16 microsatellite loci and p-values for one-tailed Wilcoxon tests for evidence of a genetic bottleneck. Diversity metrics are as follows: n—number individuals genotyped; A—mean number alleles per locus; A_R—mean allelic richness; H_{exp}—expected heterozygosity, H_{obs}—observed heterozygosity.

Population	n	A	A_R	H_{exp}	H_{obs}	Bottleneck p-value
Honey Creek	44	6.56	5.78	0.669	0.653	0.490
Snyder Creek	19	6.88	6.88	0.680	0.668	0.798
Deep Creek	26	8.25	7.70	0.775	0.811	0.935
Twentymile Creek	75	6.94	5.84	0.739	0.755	<0.001
Summer Lake Irrigation Canal	68	6.75	5.88	0.690	0.678	0.735

3.2 | Genetic population assignments

Nearly all baseline individuals were assigned to the tributary they were collected from in the leave-one-out tests. Exceptions were two fish collected in Deep Creek assigned to Honey Creek and one fish collected in Twentymile Creek assigned to Deep Creek (Table 5). Likelihood scores for the three fish assigned to a tributary they were not collected from ranged from 85.8 to 99.9 and likelihood scores for baseline individuals assigned to their collection location were all greater than 99.5 (Appendix S3). Nearly all fish collected in Crump Lake between 2006 and 2012 were assigned to Deep Creek (Table 6). The exceptions were one individual collected in 2008 that was assigned to Honey Creek and one individual collected in 2008 that was assigned to Snyder Creek. Likelihood scores for genetic assignments of fish collected in Crump Lake ranged from 80.9 to 100.0 and all but two fish had likelihood scores greater than 97.0 (Appendix S3). The majority (144 of 232) of fish collected in Hart Lake were assigned to Deep Creek; however, a number of fish from Hart Lake (88 of 232) were also assigned to Honey Creek (Table 6). Most fish collected from Hart Lake in 2006 and 2008 were assigned to Deep Creek, but in 2012, nearly equal proportions of individuals collected in Hart Lake were assigned to Deep and Honey creeks (Table 6). Likelihood scores for genetic assignments of fish collected in Hart Lake ranged from 58.6 to 100.0 and approximately 88% of genetic assignments of fish collected in Hart Lake had likelihood scores of 96.0 or greater (Appendix S3).

4 | DISCUSSION

4.1 | Genetic variation within and among tributary populations

Information regarding the level of genetic variation within populations is important when developing management and conservation

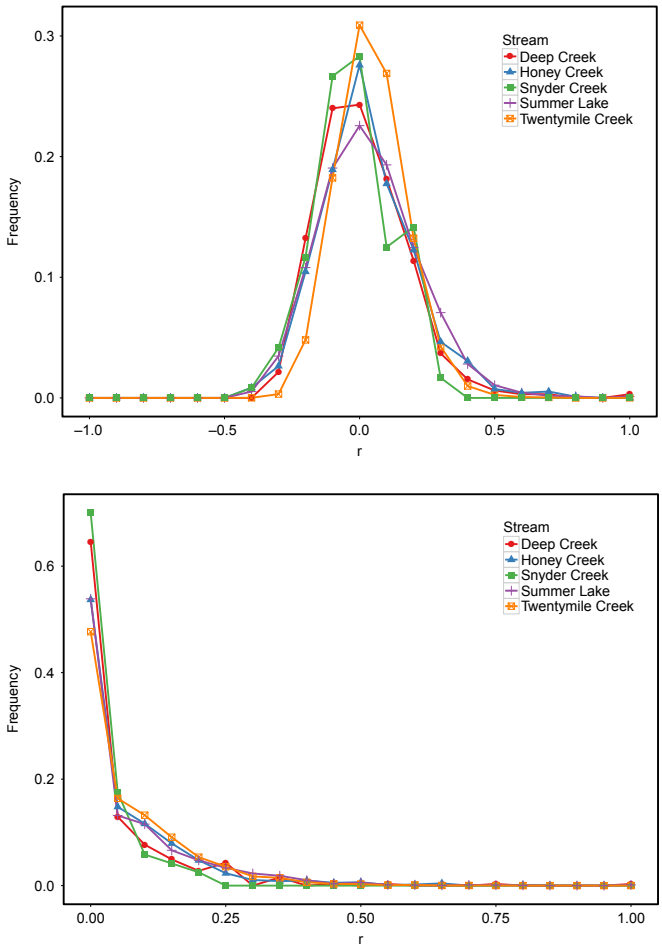

FIGURE 2 Distribution of pairwise relatedness values for five Warner sucker populations. The top figure represents Queller and Goodnight (1989) pairwise relatedness (*r*) values and the bottom figure represents Kalinowski et al. (2006) pairwise *r* values

TABLE 4 Genetic variation among Warner sucker populations. Values below the diagonal represent pairwise F_{ST} values and values above the diagonal represent the number of loci (out of 16) that showed significant allele frequency differences among populations.

	Honey Creek	Snyder Creek	Deep Creek	Twentymile Creek	Summer Lake Irrigation Canal
Honey Creek	–	13	14	16	16
Snyder Creek	0.092	–	16	16	16
Deep Creek	0.082	0.114	–	15	15
Twentymile Creek	0.180	0.175	0.095	–	16
Summer Lake Irrigation Canal	0.212	0.211	0.104	0.139	–

strategies for threatened and endangered species. Populations with low levels of genetic diversity may be unable to adapt to changing environmental conditions and may face an increased risk of extinction (Allendorf & Luikart, 2007; Reed & Frankham, 2003). As a result, these populations may represent higher priorities for conservation. The ability to adapt to a changing environment is especially important for species such as the Warner sucker that inhabit arid landscapes where water temperatures, stream flows and habitat availability can fluctuate widely from year to year, particularly in the face of anthropogenic disturbance and a changing climate (Jaeger, Olden, & Pelland, 2014; Sada & Vinyard, 2002). Given that Warner sucker habitat has been extensively fragmented and populations have presumably declined from historical levels, we might expect relatively low estimates of genetic variation based on population genetic theory (Allendorf & Luikart, 2007). However, within-population estimates of genetic variation appeared moderate and were relatively consistent among populations, suggesting that no Warner sucker population faces any increased risks associated with low genetic variation. Furthermore, estimates of heterozygosity we report (roughly 0.6–0.8) were similar to those observed for a number of different Catostomid taxa from across North America (Bessert & Orti, 2008; Darden & Tarpey, 2014; Hopken, Douglas, & Douglas, 2013; Lippe, Dumont, & Bernatchez, 2006; Reid, Wilson, Carl, et al., 2008; Reid, Wilson, Mandrak, et al., 2008). Warner suckers are a relatively long-lived species with overlapping generations (adults can live upwards of 20 years; White et al., 1991), and these life history traits have likely helped buffer populations against severe declines in genetic variation. Furthermore, periodic connectivity among populations may occur during winter and

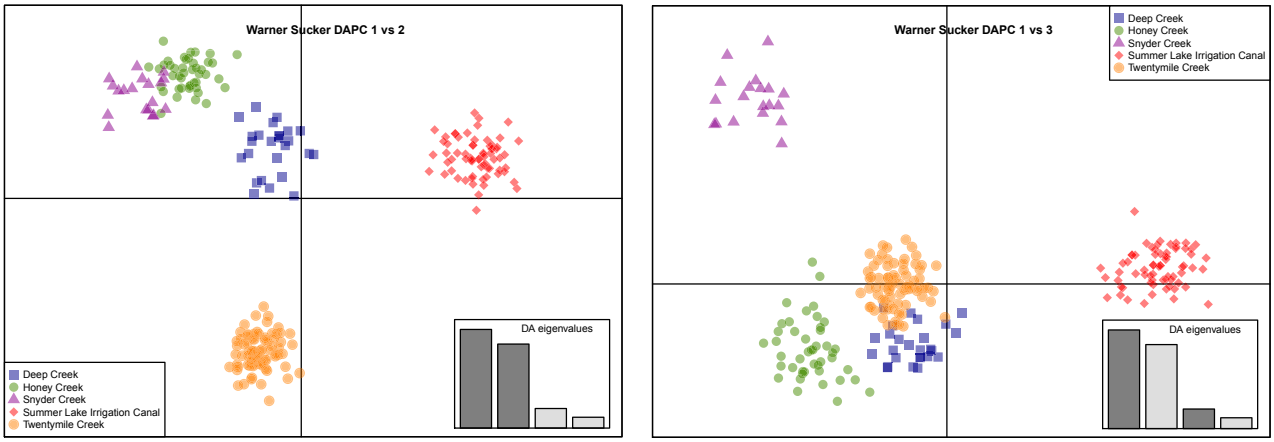

FIGURE 3 Discriminant analysis of principal components (DAPC) based on allele frequencies of five Warner sucker populations. Each point on the graph represents an individual in our analysis and symbols and shading denote the different populations. The plot on the left represents the first and second principal components and the plot on the right represents the first and third principal components

TABLE 5 Proportion of individuals assigned to each baseline population in leave-one-out genetic assignment tests. The first column represents the collection location and subsequent columns represent assignment locations. Bold text represents the proportion of individuals that were assigned to the tributary population they were collected from.

	Assigned to			
Collection site	Honey Creek	Snyder Creek	Deep Creek	Twentymile Creek
Honey Creek	**1.000**	0.000	0.000	0.000
Snyder Creek	0.000	**1.000**	0.000	0.000
Deep Creek	0.077	0.000	**0.923**	0.000
Twentymile Creek	0.000	0.000	0.013	**0.987**

TABLE 6 Genetic population assignments of Warner suckers collected in Hart and Crump lakes from 2006 to 2012.

	Assigned to		
Collection site	Deep Creek	Honey Creek	Snyder Creek
Crump Lake 2006	2		
Crump Lake 2008	24	1	1
Crump Lake 2010	30		
Crump Lake 2012	34		
Hart Lake 2006	8	5	
Hart Lake 2008	60	14	
Hart Lake 2012	76	69	

early spring high flow events when irrigators are not operating the diversions. These periodic flow events may facilitate downstream migration into the lakes and may be the current mechanism by which substantial recruitment into the lakes occurs (Scheerer et al., in press). The pattern of retaining genetic variation despite declines in abundance has been documented in many other long-lived desert fishes and Catostomid species (Darden & Tarpey, 2014; Dowling, Marsh, Kelsen, & Tibbets, 2005; Finger & May, 2015; Garrigan, Marsh, & Dowling, 2002; Lippe et al., 2006).

One population, Twentymile Creek, did show evidence of a recent genetic bottleneck. Recent surveys showed that Twentymile Creek had the greatest estimates of abundance and density compared to all other tributary populations (Scheerer et al., in press), suggesting that although abundance remains relatively high, there has been a loss of genetic diversity in Twentymile Creek in recent generations. One key difference between Twentymile Creek and the other tributary populations is that that Warner suckers in Twentymile Creek are isolated from the Warner Lakes by an extensive network of irrigation canals and diversions, whereas other populations still have some degree of connectivity to the Warner Lakes and to each other. Population isolation and reduced gene flow has presumably led to a loss of genetic variation in Twentymile Creek. The loss of genetic variation has been repeatedly documented in isolated fish populations (DeHaan, Brenkman, et al., 2011; Meldgaard, Nielsen, & Loeschcke, 2003; Neville, Dunham, & Peacock, 2006; Wofford, Gresswell, & Banks, 2005; Yamamoto, Morita, Koizumi, & Maekawa, 2004), including other species of suckers (Bessert & Orti, 2008; Reid, Wilson, Mandrak, et al., 2008).

Translocations and population introductions are often used to establish refugial populations of threatened and endangered fishes in the event that natural populations are extirpated (George et al., 2009). Refugial populations are often founded with low numbers of individuals, which may result in a loss of genetic variation and/or genetic bottleneck (Finger, Parmenter, & May, 2013; Mock et al., 2004; Puckett et al., 2014; Stephen et al., 2005). The refugial population in the Summer Lake WMA was established during a drought in 1991 when Hart and Crump lakes completely desiccated (White et al., 1991). Although this population has not been used to re-establish extirpated sucker populations, information on the level of genetic diversity in this population is important if the population is to be maintained and potentially utilised in the future. Although the number of individuals initially introduced into Summer Lake WMA is known, it is unclear how many of those individuals spawned before they were removed (White et al., 1991). Estimates of genetic diversity did not differ significantly

between Summer Lake and three of the four natural populations, and the Summer Lake population did not show evidence of a genetic bottleneck. Given the skewed sex ratio of the founders and the possibility that only a small subset of these individuals spawned, we might expect to observe increased pairwise relatedness in the Summer Lake population. This was not the case, however; pairwise estimates of relatedness did not differ between the introduced population and the four natural populations. Collectively, these data suggest that the amount of genetic variation in the Summer Lake population is similar to genetic variation in natural populations. It should be noted, however, that the Summer Lake population remains isolated and will lose variation via genetic drift in future generations. Managers may consider facilitated gene flow via periodic transfers of individuals from the naturally occurring populations to minimise this risk (Frankham, 2015; Tallmon, Luikart, & Waples, 2004).

Previous analyses have shown that the spatial scale at which genetic variation is partitioned varies widely among Catostomids. Whereas some species exhibit significant genetic variation within large river basins (Dowling, Saltzgiver, & Marsh, 2012; McPhee, Osborne, & Turner, 2008; Turner, Osborne, McPhee, & Kruse, 2015), within individual rivers (Reid, Wilson, Carl, et al., 2008) or even within individual tributaries (Smith, Reid, Godfrey, & Ardren, 2011), other species exhibit relatively low levels of genetic variation across large watersheds (Reid, Wilson, Mandrak, et al., 2008). Warner suckers exhibited significant genetic variation among tributaries and even within tributaries (e.g. Honey and Snyder creeks), indicating that each of the four tributaries sampled represents a genetically distinct spawning population. Although genetic variation among stream-dwelling freshwater fish populations is often a function of geographic distance and/or stream hierarchy (DeHaan, Bernall, et al., 2011; Hopken et al., 2013; Taylor, Gow, Witt, & Zemlak, 2011; Turner et al., 2015), our data suggest that habitat and population characteristics shape genetic variation among Warner sucker populations. For example, although Deep and Twentymile creeks are geographically proximate tributaries both within the Crump Lake drainage, F_{ST} values and the DAPC plot showed that Deep Creek was more similar to Honey Creek (a tributary to Hart Lake) than Twentymile Creek. As noted above, habitat in Twentymile Creek is extensively fragmented and the population is isolated from the Warner Lakes and other populations, whereas habitat in Deep and Honey creeks remains connected to the Warner Lakes and both populations maintain a migratory life history component.

4.2 | Genetic population assignments

Life history diversity in fishes helps maintain population viability by spreading threats and risks among different components of the population (Greene et al., 2010; Hilborn et al., 2003; Schindler et al., 2010). Warner suckers have evolved two distinct life history types and historically interactions between the life history types helped to maintain population viability in a changing, arid environment. Prior to the current study, virtually no information existed regarding the tributary origins of Warner suckers in Hart and Crump Lakes and the interactions between the two life history types. Genetic assignment data showed that all but two fish collected in Crump Lake and nearly two-thirds of fish collected in Hart Lake originated in Deep Creek. These data indicate that Deep Creek is the primary population producing migratory, lake-type individuals. Presumably, large, highly fecund individuals in the lakes can make a substantial demographic contribution to stream populations (Blueweiss et al., 1978; Kamler, 2005), they provide a means for genetic exchange among populations, and they act as a demographic and genetic reserve in case of stochastic events that negatively impact stream populations (e.g. fires, landslides). In order to maintain the lake-type life history for Warner suckers in the near term, it will be important to protect the population in Deep Creek and maintain the connection between habitat in Deep Creek and the Warner Lakes.

The relatively low abundance of fish from Honey, Snyder and Twentymile creeks in the Warner Lakes is presumably due to the presence of numerous barriers in the lower reaches of these tributaries that limit connectivity between these populations and habitat in the lakes. Conservation efforts for Warner suckers have recently been focused on improving connectivity between stream and lake habitats and improving fish passage at existing irrigation structures in the tributaries (Scheerer, Peterson, & Clements, 2014). The low abundance of Honey, Snyder and Twentymile creek origin fish in the Warner Lakes suggests that the populations in these creeks would benefit from management actions that aim to increase habitat connectivity and fish passage for Warner suckers. Improved connectivity between stream and lake habitats and among populations is essential for maintaining a functioning metapopulation of Warner suckers, as specified in the species recovery plan (USFWS 1998). The 2012 sample from Hart Lake did contain nearly equal proportions of fish from Honey and Deep creeks. These samples were collected following a year with increased precipitation and stream flows in the Warner Basin, which resulted in less water being diverted from the streams for irrigation purposes. The absence of Twentymile Creek origin fish in the 2012 samples from the lakes, despite the high flow events the previous winter, may also indicate loss of the migratory life history from this population. These data highlight the fact that maintaining stream flows in the Warner Basin is important for maintaining connections between the streams and the lakes and for maintaining the life history diversity of Warner suckers. This will likely become increasingly challenging in the future assuming that stream flow regimes change as predicted by climate change scenarios for the Great Basin (Chambers, 2008; Pilliod, Arkle, Robertson, Murphy, & Funk, 2015) and anthropogenic demands for water in the Great Basin increase (Sada & Vinyard, 2002).

Three of the 164 baseline individuals were not assigned to their collection location during leave-one-out assignment tests. These individuals may represent either migrants among baseline populations or they may have simply been mislabelled. All three of these fish were collected on a day when sampling crews only visited a single collection site, making it unlikely that the wrong sampling location was recorded for these fish. It is more likely that these individuals represent migratory fish that were collected in a tributary they did not originate from or that they are the progeny of migratory fish. Data presented here demonstrate that Warner suckers do migrate between Hart and

Crump lakes and PIT tag data have documented movement from the lakes into the tributaries (Scheerer et al., in press). Deep Creek and Honey Creek are hydrologically connected via the Warner Lakes and migration is plausible. Connections between Twentymile Creek and Deep Creek presumably only occur during very high flows making migration between these two tributaries less likely, although we cannot completely eliminate the possibility. Furthermore, Warner suckers are relatively long-lived and the individual collected in Twentymile Creek that assigned to Deep Creek (or perhaps one or both of its parents) could have migrated several years prior to collection when flows were much more favourable.

5 | CONCLUSIONS

Historically, life history diversity allowed Warner suckers to persist in an arid environment where habitat quality and availability routinely fluctuated. Declines in the species' abundance due to habitat fragmentation and degradation as well as the proliferation of non-native piscivorous fishes in the Warner Lakes during the last century led to the species being listed as a "threatened" species and the need for increased conservation measures. One of the primary recovery goals for Warner suckers is the maintenance of a self-sustaining metapopulation (USFWS 1985). This implies that opportunities exist for migration and gene flow among populations (Hanski & Gilpin, 1991). However, genetic data presented here corroborate earlier findings based on PIT tags and suggest that presently there is very limited movement and genetic exchange among populations (Scheerer et al., in press). Although the population in Deep Creek apparently maintains a migratory life history component, the low abundance of fish from other tributaries in the Warner Lakes highlights the need for increased connectivity between the other tributaries and the Warner Lakes. Recently, recovery actions have focused on modifying or replacing ageing irrigation structures with structures that have fishways and screened diversions designed to reduce the negative impacts on Warner suckers (Scheerer, Peterson, & Clements, 2015; Scheerer et al., 2014). Additionally, management agencies may consider reducing the abundance of piscivorous fishes in the lakes to increase sucker abundance and increase the number of potential spawners that have access to the tributaries. Challenges associated with maintaining connectivity between stream and lake habitats will presumably increase in the future as changes in climate are predicted to result in reduced spring flows in the Great Basin (Chambers, 2008; Pilliod et al., 2015), which will presumably increase habitat fragmentation and lake desiccation for Warner suckers.

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